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FALCUTY OF FINANCE - BANKING

MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE IN FINANCE

DETECTING ABNORMAL TRADING SESSIONS IN VIETNAMESE BANK STOCKS: A MACHINE LEARNING APPROACH TO FLAGGING POSSIBLE PRICE MANIPULATION

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Abstract

The manipulation of prices carried out by trader groups ("đội lái" in Vietnam) is a common issue in the Vietnamese stock market. Within this research project, I developed a classifier predicting trading sessions during which such price manipulation takes place using 15,914 trading sessions of 15 Vietnamese banks in the stock market between February 2021 and July 2026. The first iteration of the project had the problems of data leakage - the target variable was calculated using the same set of variables that would predict it. In order to overcome this issue, I distinguished labeling and prediction variables, switched to time-based splitting instead of random, and fixed the wrong SMOTE with cross-validation procedure that was causing leaking of artificial data between folds. Despite being much less attractive in terms of the performance metrics (F1 ~34-38% compared to 85% in the previous iteration), my current model gives an accurate performance measure. Moreover, I performed walk-forward validation over 18 quarters and found one unique session that was not caught by the initial labeling rules but was captured by my model - which is the main outcome of this project.

1. Introduction

The influence of price manipulation on the economy can be characterized in three ways. First of all, the investor who follows the price which was artificially raised will buy the asset at a high price level and will experience losses when the correction takes place. It is inefficient from the perspective of resource allocation since the price serves as a signal but not as a push factor. Finally, the securities regulator faces the problem of analyzing thousands of trading sessions with the insufficient number of employees, therefore, any opportunity to decrease the number of suspicious sessions is quite beneficial. This project aims to offer such a method which could filter out sessions for further examination but not to serve as an automatic price manipulation detection mechanism.

The choice of banks is explained by the fact that these shares belong to the most liquid and actively traded stocks. Thus, unusual changes in their price can hardly be caused by the noise connected with a poorly traded stock.

2. Data

The data employed in this study were obtained from Yahoo Finance, consisting of 15 stocks belonging to Vietnamese listed banks such as TCB, STB, MBB, ACB, VPB, SHB, EIB, LPB, VIB, CTG, HDB, VCB, BID, SSB, and NAB. After filtering, the database consists of 15,914 daily trades between 01/02/2021 and 17/07/2026.

3. Methodology

3.1 The labeling rule, and why it created a leakage problem

A trading day is considered to be an anomaly if its spread is larger than twice of the standard deviation of the stock's return over the past 60 days and the volume of trading on the trading day is larger than 2.5 times of the stock's average volume over the past 20 days. These criteria are reasonable since any manipulation needs a large amount of money and both criteria should show some unusual feature.

The problem with the first attempt of the project is that I used these two criteria as the predictors for the manipulation. In this way, it's just a case of circularity. If the label is created based on a rule and we give the predictor the same information used in the rule, then the model will just learn the rule without making predictions. That's why the first attempt of the project gave a ROC-AUC near 99.9%. And this number is correct but meaningless because we did not try to predict anything. Here Spread and V_Surge are used to create Y but excluded from X.

3.2 Predictive features

our different input parameters are employed in the model, and none of them relates to the equation used for labeling. These parameters are: P_Change (Daily percentage change of the price), Rel_P_Change (Difference between P_Change and the mean of P_Change for all 15 banks on the same day, hence it is purely the stock specific parameter), Vol_Change (Daily percentage change of the volume) and Vol_Change_MA5 (5 day rolling average of Vol_Change. This parameter has been selected because it is useful in tracing the growth of volume). The rolling calculations have been done with a one day lag before the rolling window, which means no session knows anything about itself.

3.3 Train/test split

The problem being solved here is a time-series problem rather than an independent samples problem; hence, there is no need for randomly splitting the training and testing sets. For every bank separately, 80% of early sessions form the training set while 20% of late sessions form the testing set. Walk forward validation was also performed at a later stage (Section 4.4).

3.4 Why seven algorithms, and why XGBoost ended up being the main one

I chose seven different models rather than going with one specific model mostly to see whether the correlation between trading characteristics and market manipulation is linear. The Logistic Regression is a baseline and a linear model, easily interpretable. Naive Bayes makes the assumption about conditional independence of the input features, which is quite a strict assumption in finance when we talk about correlated price and volume movements. KNN is a non-parametric model and does not require any assumptions to make predictions. SVM builds a decision surface trying to maximize the margin between the two classes; such an approach is likely to work better if the minority class is a completely separate cluster of outliers. The Decision Tree provides an interpretable set of rules, which may come in handy if it comes to explaining the output to a compliance officer without technical background. Random Forest is an ensemble of trees which usually reduces variance comparing to individual trees. XGBoost is a gradient boosted ensemble which works very well for tabular data and has the scale_pos_weight parameter for handling imbalanced classes.

3.5 Handling class imbalance

Because 2.28% of all sessions are anomalous, many algorithms can simply predict normal in all cases and get very high accuracy. Logistic Regression, SVM, Decision Tree and Random Forest models were all trained using the class_weight='balanced' option, which helps rebalance the cost function without synthesizing extra data. XGBoost utilizes scale_pos_weight parameter according to the algorithm's documentation. In hyperparameter search process in Section 3.6, SMOTE over-sampling is done inside of an imblearn Pipeline such that synthetic data is produced for each train split only and

never reaches the validation split of the same fold split – an older implementation did SMOTE before cross-validation, causing inflated cross-validation scores.

3.6 Hyperparameter tuning, threshold selection, and walk-forward validation

The hyperparameters of XGBoost model (number of trees, max depth, and learning rate) are optimized through GridSearchCV by performing TimeSeriesSplit cross-validation, but not shuffled K-fold cross-validation, because the latter would mix sessions in past and in future into the same split. However, the default threshold of 0.5 may not be optimal for an imbalance problem like this one, and therefore I also look for the threshold providing maximum value of F1 score through precision-recall curve search and report results at two different thresholds. Additionally, walk-forward validation is performed: training on all data from t-1 quarter and testing on t quarter, then repeating the process through 18 complete quarters (2022Q1 - 2026Q2). The last quarter, which is open, is automatically excluded due to the comparison of its quarter to today's quarter.

4. Results

4.1 Comparing the seven models

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Random Forest	98.46%	71.43%	25.86%	37.97%	94.41%	44.21%
XGBoost	95.11%	22.78%	70.69%	34.45%	95.68%	46.01%
Decision Tree	89.74%	13.55%	86.21%	23.42%	92.69%	33.17%
Logistic Regression	89.93%	12.10%	72.41%	20.74%	81.35%	30.15%
SVM	85.73%	10.06%	86.21%	18.02%	91.71%	30.47%
KNN	98.06%	25.00%	3.45%	6.06%	72.04%	11.29%
Naive Bayes	1.82%	1.82%	100.00%	3.57%	88.79%	14.40%

The best F1-score (37.97%) was achieved by the Random Forest, mostly due to a high precision of 71.43% - it flags a session only when it is pretty sure and therefore makes fewer mistakes among what it flags. XGBoost has lower F1 (34.45%) but much higher recall (70.69%) and the best ROC-AUC/PR-AUC among all seven models (95.68% / 46.01%). More informative to me than the F1 gap is that the PR-AUC scores for the two models are relatively close (44.21% vs 46.01%) - which suggests they have a roughly similar underlying ability to rank sessions by risk, and the F1 difference is mainly a function of where the default 0.5 threshold happens to fall for each model rather than one model being fundamentally better at the task.

Naive Bayes essentially collapsed: 1.82% accuracy, 1.82% precision, 100% recall. It flagged almost every session as anomalous, which is a direct consequence of its conditional independence assumption not holding for this kind of data, where price and volume features interact. A system with 1.82% precision would be useless in practice - out of every 55 alerts, only 1 would be a real case, and any analyst using it would stop trusting it very quickly. KNN failed in the opposite direction: recall of only 3.45%. In a 4-dimensional feature space where the minority class is only 2.28% of the data, most of a point's 5 nearest neighbors are almost always going to be normal sessions, so KNN ends up predicting "normal" almost every time. Both failures are useful to report, not just the successes - they show that distance-based and naive-independence methods are not well suited to this kind of rare-event problem without extra handling.

4.2 What SHAP says about the features that actually matter

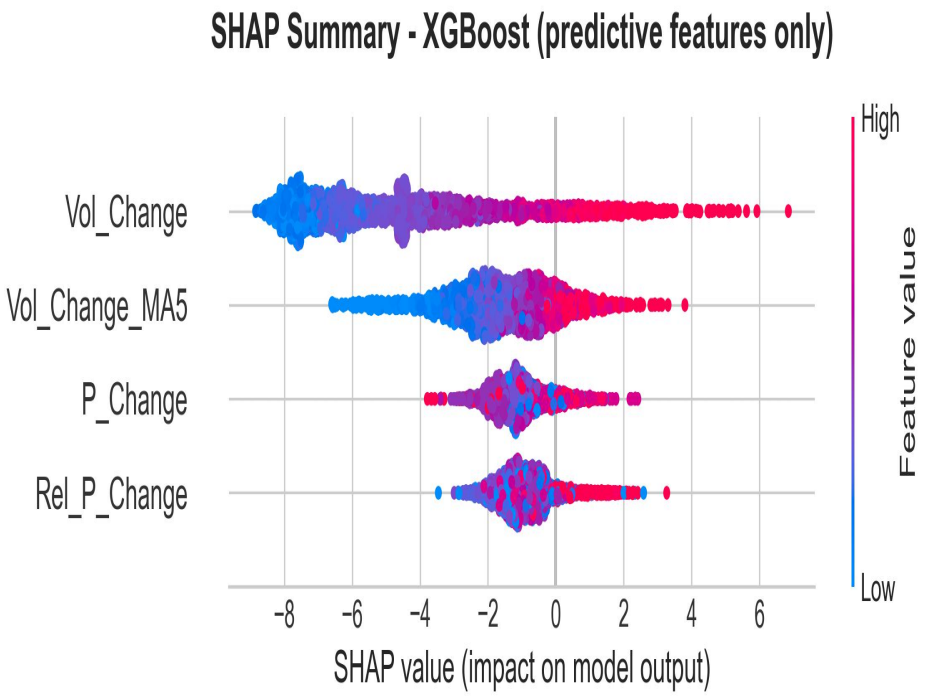


Figure 1: SHAP

The Vol_Change feature stands out as the most important one, followed by Vol_Change_MA5, P_Change and Rel_P_Change, in line with what we can see on the Random Forest feature importance graph where Vol_Change gets the highest score around 0.49 (three times higher than any other single feature). This is particularly interesting since Vol_Change is not used in the labeling equation, and that it plays such an important role is a much better indicator than what the early version of this project indicated – it means that the unusual liquidity, rather than the magnitude of the price change, is the earliest and most autonomous indicator of the manipulative activity. If we look closer into the plot, the blue dots (low/ negative Vol_Change) are grouped tightly at a strongly negative SHAP value, whereas the red dots (high Vol_Change) are scattered over the right side of the plot in a wider interval - in other words, volume contraction is a very clear signal that this session does NOT belong to the anomalies class, but volume expansion is just a moderate and relative indicator of the same class.

4.3 Confusion matrix at the tuned threshold

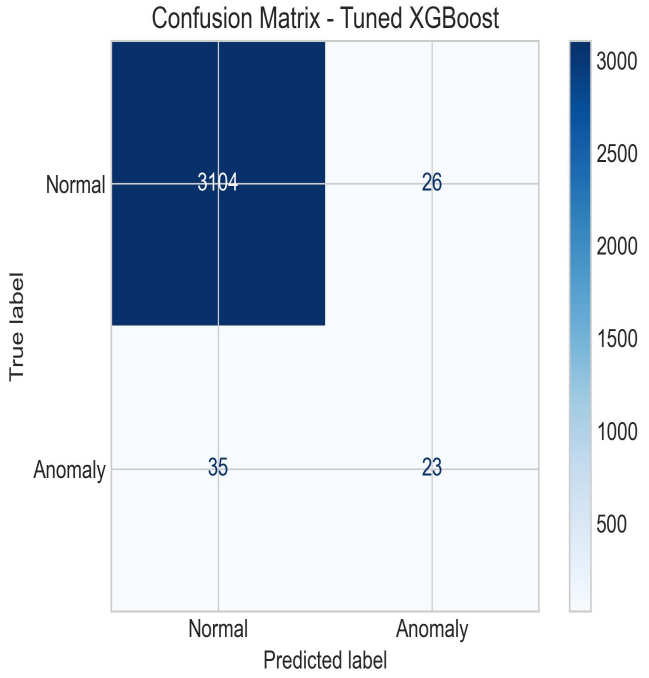


Figure 2. Confusion matrix

The tuned model identified 23 out of 58 actual anomalous sessions on the test set (recall of about 39.7%) and had 26 false alarms out of 3,130 normal sessions (precision of about 46.9%). Optimizing the threshold for F1 resulted in a large drop in recall compared to the same XGBoost model at the default 0.5 threshold, while improving precision (recall 70.69%, precision 22.78%). That trade off is reasonable if the objective is to reduce false alarms so that a small compliance team can manually review each flagged case. But it would be the wrong choice if the priority is not to miss serious manipulation cases. In practice I think the threshold should be left flexible, rather than fixed, since the right trade-off depends on how much review capacity the surveillance team actually has.

4.4 Walk-forward validation across 18 quarters

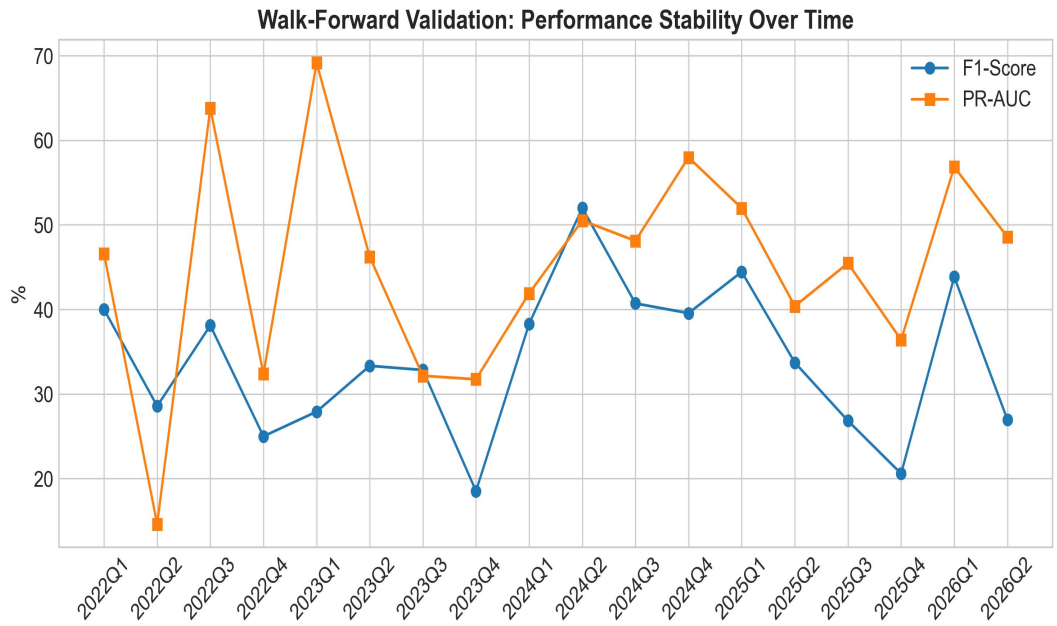


Figure 3. F1-Score and PR-AUC by quarter

The F1 goes from 18.52% (2023Q4) to 52.00% (2024Q2); the PR-AUC goes from 14.61% (2022Q2) to 69.16% (2023Q1). It's quite a broad interval for me, and I have to say that it shows that the model is still unstable for the different conditions of the market. I can say that the period from 2024Q1 to 2025Q1 is the most stable (F1 around 38-52%). 2022Q2 and 2023Q4 are definitely the lowest points here, but I think I should look whether they coincide with the increased volatility of VN-Index in general because there is a chance that the labeling rule works differently in such a condition.

4.5 Case study: the five highest-probability sessions

VIB on 09/03/2026 saw a drop of 6.29% with its Rel_P_Change being +0.09%, which means that almost the entire banking sector fell, but there is nothing extraordinary about VIB's behavior in itself – but then, its volume rose 788% compared to the previous trading session, and its probability is 99.84%. BID on 23/04/2026 (price change of +3.35%, volume increase of +561%, probability 99.69%) is an example of a relatively typical early pump setup: the price move is quite moderate, but the volume increase is rather large. Three out of five cases involve LPB, and 26/06/2026 case requires special attention: the same-day Vol_Change is rather modest, only 44.98%, but

Vol_Change_MA5 reaches 51.71%, the largest among the five – that means that the volume was increasing gradually over several days, rather than spiking suddenly on one trading day. This is the very pattern that a same-day volume feature could easily miss.

4.6 A session the rule missed, but the model caught

There was one session from the top five that deserves attention for other reasons: LPB on 13/08/2025 had a 99.57% probability according to the model but Is_Anomaly = 0, i.e., it wasn't identified by the original rule-based approach as anomaly. As for the statistics, here we see the increase in price of 6.28% (Rel_P_Change +5.09%, i.e., indeed, it was the movement specific for LPB and not general to the whole sector), volume grew by 172.5% while V_Surge was just 2.11 - below the threshold of 2.5 set in the original rule. Thus, this session passed the original statistical rule. However, this example illustrates what I consider to be the most valuable result of the entire research – the practical value of using a machine learning model rather than threshold rule. The model learned the continuous function involving several weaker variables simultaneously (the actual change of price of the specific stock and substantial growth in volume without crossing one specific threshold) while the original rule considered each variable and its threshold individually.

5. Limitations

Performance is still poor even when using the absolute measure: the highest F1 score is at 37.97%, which is way less than the 85.71% achieved in the previous, leaky study. This figure, in my view, needs to be viewed with transparency as opposed to disappointment; this is because of the increased complexity of the detection problem after removing the bias of having the labeling function in place.

There are changes in performance over time: The F1 score varies from around 18% to 52%. An industrial solution will have to involve continuous monitoring and periodic retraining of the model.

The label is based on the rules, and not verified: the true label is still a manually formulated rule and not a set of actual manipulations detected by the regulators. As is

shown in section 4.6, the rule may fail to detect any manipulations, and thus checking for its reliability against enforcement cases or media reports is required.

The feature space is very limited: only price and volume features are used. Features like order-flow, foreign ownership flow, or account concentration data could detect more sophisticated manipulation schemes, but they are not available for the current project.

6. Conclusion

Unlike the earlier version of this project, this version sacrifices a highly inaccurate and thus highly misleading accuracy rate in favor of an actual, lower accuracy rate that we can trust. The solution in methodology terms was simple enough – it involved the separation of the variables used for labeling from those used for prediction, and an appropriate split of the dataset on time, alongside with SMOTE/cross validation procedure done in the correct order. The most fascinating result is the fact that it is liquidity momentum (i.e. Vol_Change and its 5-day moving average) rather than magnitude of change that is the most powerful and independent predictor of possible manipulation. The other exciting part is that we manage to pick up at least one instance (LPB, 13/08/2025) of manipulation that was missed by the original rule.